



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION



Project Title

SMART CAMPUS LOCATOR SYSTEM

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Introduction

In big colleges, students and visitors often face problems finding classrooms, labs, admin block, canteen, library, and other places inside the campus. New students mostly get confused and waste time asking people for directions.

Our project “Smart Campus Locator System” is made to solve this problem. This system helps users find locations easily with smart navigation and useful campus features.

The project will make campus navigation simple, fast, and user friendly.

Problem Statement

Students and visitors face many difficulties while moving inside the college campus.

Some common problems are:

- Students cannot find classrooms quickly*
- Visitors get confused inside campus*
- Time gets wasted searching for locations*
- No smart digital guidance system*
- Hard to find empty classrooms for self study*
- Students miss important events and notices*
- Lost items are difficult to report*
- In emergency situations students may panic*

So, there is a need for a smart campus system that can help users navigate the campus easily and provide important services in one platform.

Proposed Solution

We are developing a Smart Campus Locator System that helps students and visitors locate places inside the campus easily.

The system will include:

- *Smart navigation*
- *AI chatbot assistant*
- *Voice assistant*
- *Empty classroom finder*
- *Event and notice section*
- *Lost and found section*
- *Emergency SOS feature*

This project can work as a website or mobile application.

Objectives of the Project

The main objectives of this project are:

1. *To help students find locations easily*
2. *To reduce confusion inside campus*
3. *To save time during navigation*
4. *To provide smart campus services*
5. *To improve student experience*
6. *To create a modern digital campus system*

Features we can add

1. Smart Navigation

This feature helps users reach their destination easily.

- *Shows shortest path*
 - *Step-by-step directions*
 - *Easy campus navigation*
 - *Helpful for new students*
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2. AI Chatbot Assistant

3. Voice Assistant

4. Empty Classroom Finder

5. Lost & Found Section

6. Event & Notice Section

7. Emergency SOS

8. Interactive Campus Map

9. QR Code Navigation

QR codes can be placed around campus.

When users scan the QR code:

- *Location opens instantly*
- *Directions appear quickly*

10. Faculty Cabin Locator

Technologies we will use

Frontend

- *HTML*
- *CSS*
- *JavaScript*
- *React.js*

Backend

- *Node.js*
- *Express.js*

Database

- *MongoDB*

Tools

- *VS Code*

- *GitHub*
- *Google Maps API*
- *Firebase*

(and others also if require further)

Future Scope

In future, this project can be developed as a startup idea for colleges and universities to make campus navigation smarter and more digital .With more advanced features and real-time services, the Smart Campus Locator System can grow into a successful EdTech startup in the future.

Conclusion

The Smart Campus Locator System is a useful and modern solution for colleges and universities. It helps students and visitors find locations easily and provides smart features like chatbot support, voice assistant, empty classroom finder, and emergency SOS.

This project makes campus navigation simple, fast, and smart. It improves student experience and creates a better digital campus environment.

THANK YOU

A Step Towards Smart Education and Smart Campuses.